# Practical 8: ARIMA Modeling of AirPassengers Dataset

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## 1. Question

Consider the "AirPassengers" data from R library and write the R program for the following:

(f) Convert the data into a time series object.

(g) Plot the data to identify the dominant component.

(h) Decompose the data to observe the dominating components more clearly.

(i) Check stationarity or non-stationarity using ACF/PACF plot.

(j) Check stationarity or non-stationarity using the KPSS test.

(k) If data is non-stationary, make it stationary using an appropriate operator.

(l) Based on the dominating component, select the suitable technique to fit the data.

(m) Fit the data using the selected model and estimate the parameters of the model.

(n) Check the goodness of fit of the model.

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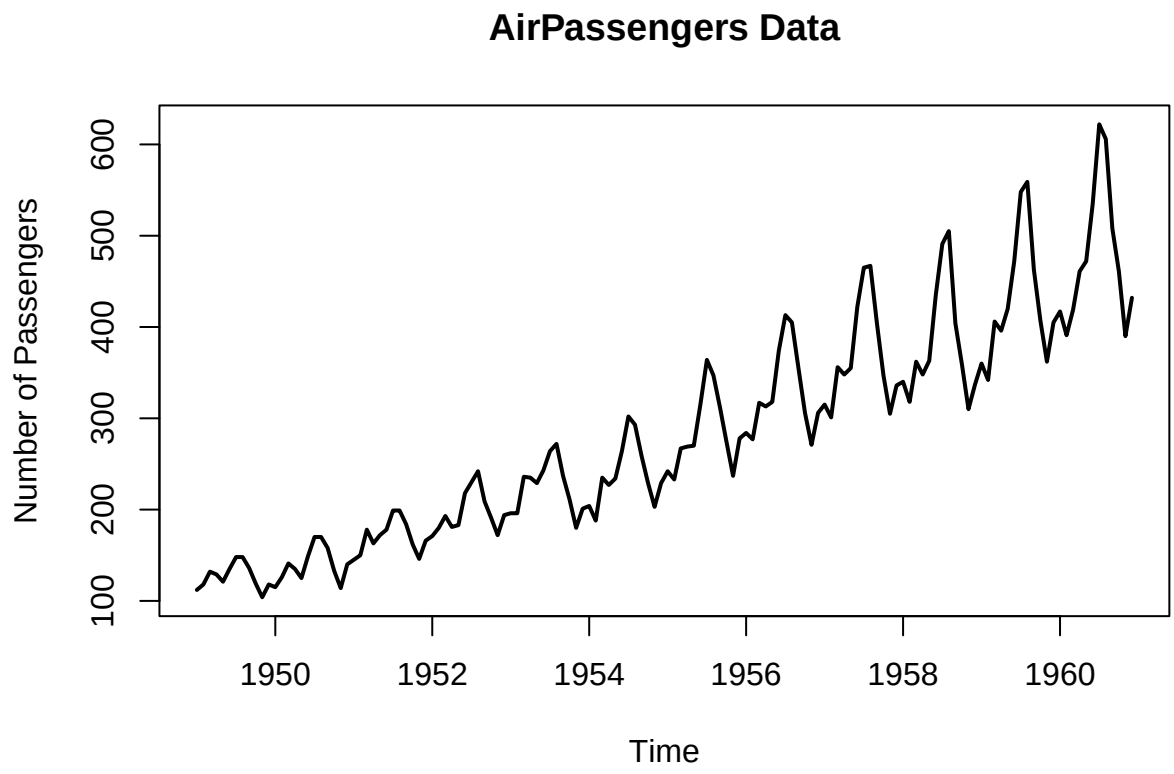
## 2. R Code (Question-wise Solution)

```r
# To delete previous history or stored variables
rm(list = ls(all = TRUE))

library(tseries)
library(forecast)
library(datasets)


# ---------------------------------------------------
# (f) Convert the data into a time series object
```

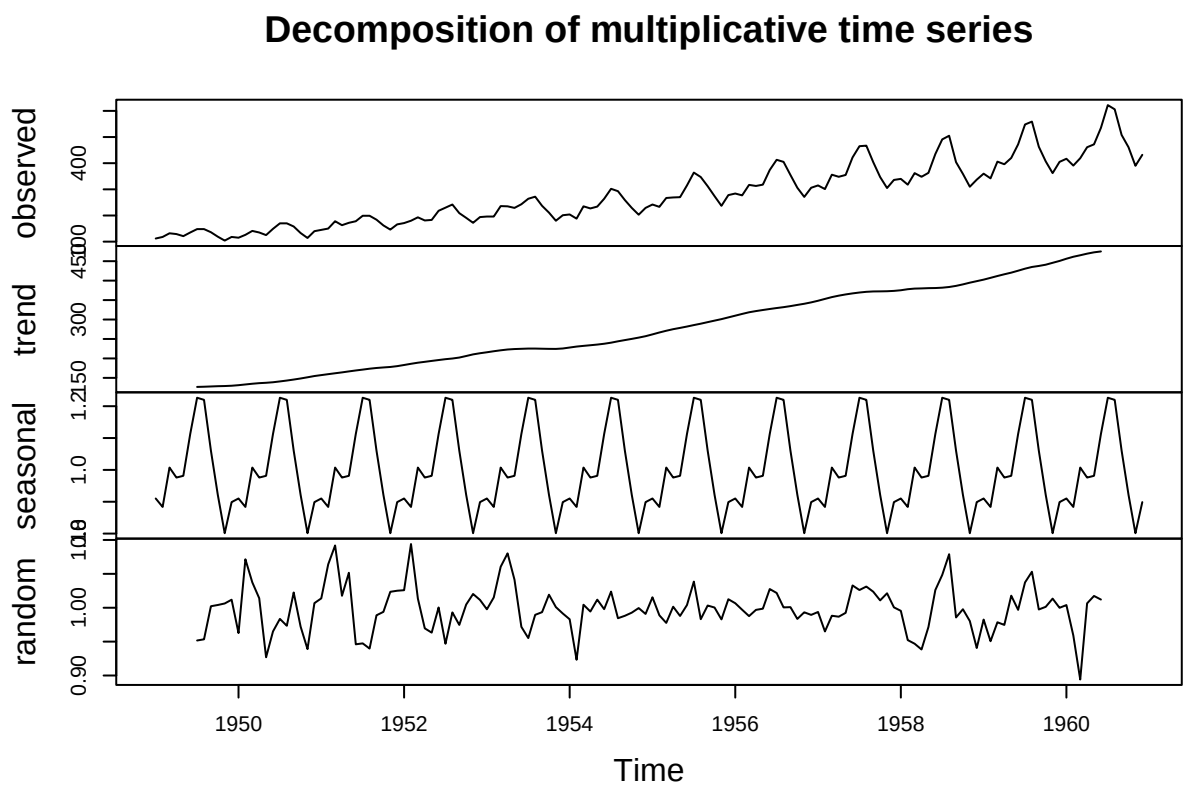
```r
# ------------------------------------------------

data(AirPassengers)

ap_data <- as.ts(AirPassengers)

# Printing the data
ap_data
```

```
##      Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec
## 1949 112 118 132 129 121 135 148 148 136 119 104 118
## 1950 115 126 141 135 125 149 170 170 158 133 114 140
## 1951 145 150 178 163 172 178 199 199 184 162 146 166
## 1952 171 180 193 181 183 218 230 242 209 191 172 194
## 1953 196 196 236 235 229 243 264 272 237 211 180 201
## 1954 204 188 235 227 234 264 302 293 259 229 203 229
## 1955 242 233 267 269 270 315 364 347 312 274 237 278
## 1956 284 277 317 313 318 374 413 405 355 306 271 306
## 1957 315 301 356 348 355 422 465 467 404 347 305 336
## 1958 340 318 362 348 363 435 491 505 404 359 310 337
## 1959 360 342 406 396 420 472 548 559 463 407 362 405
## 1960 417 391 419 461 472 535 622 606 508 461 390 432
```

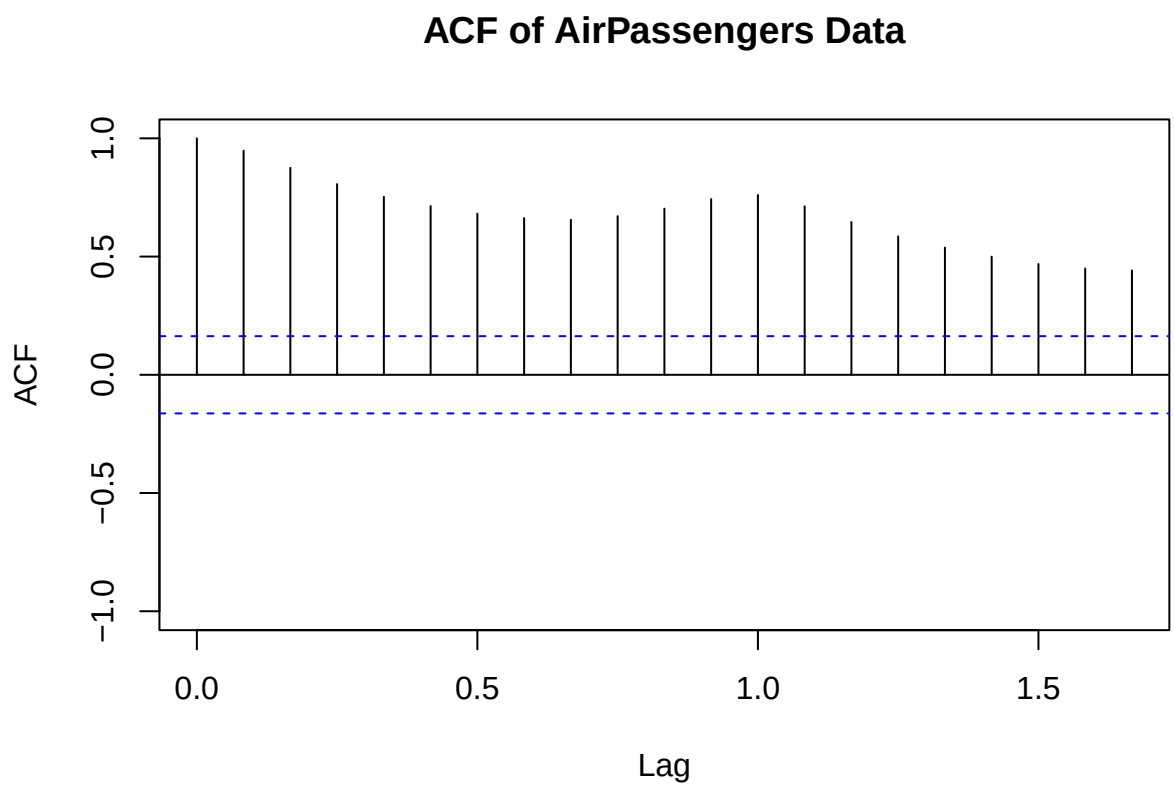
```r
# ------------------------------------------------
# (g) Plot the data to identify dominant component
# ------------------------------------------------

plot(ap_data,
     main = "AirPassengers Data",
     xlab = "Time",
     ylab = "Number of Passengers",
     lwd = 2)
```



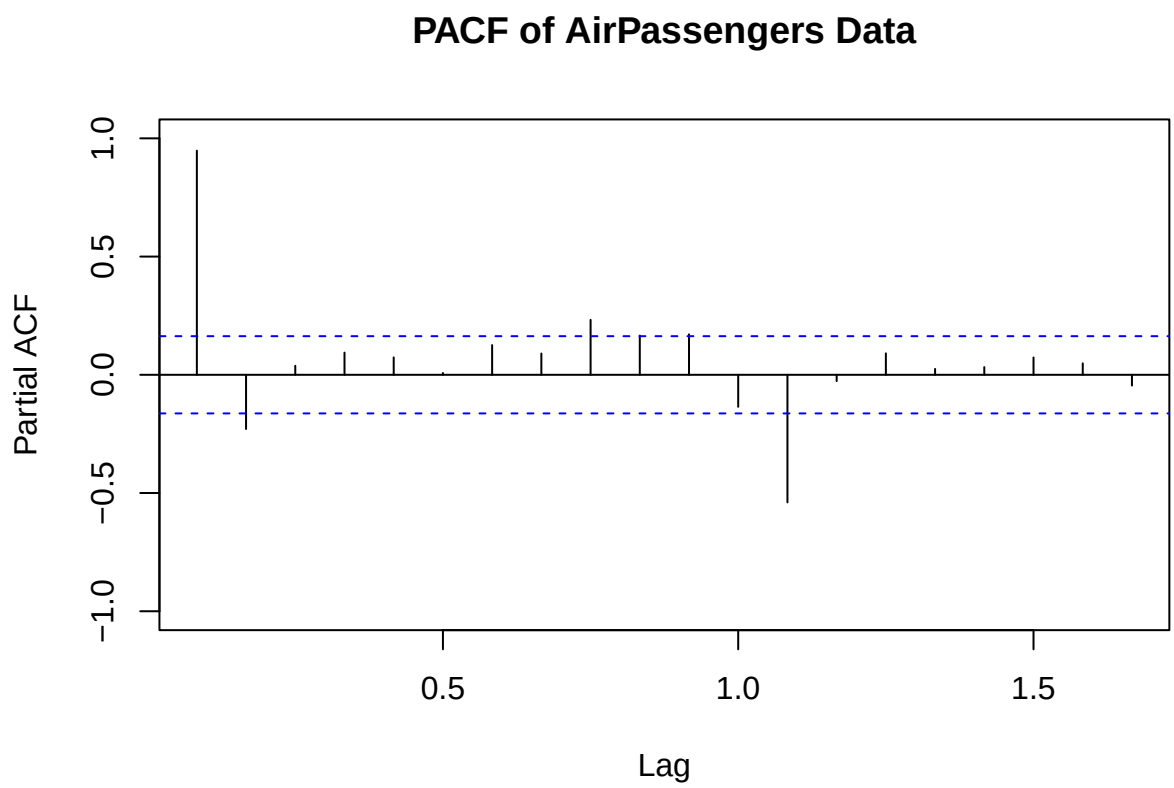

```r
# ------------------------------------------------
# (h) Decompose the data
# ------------------------------------------------

ap_decomp <- decompose(ap_data, type = "multiplicative")
plot(ap_decomp)
```




```r
# ---------------------------------------------------
# (i) Check stationarity using ACF/PACF
# ---------------------------------------------------

acf(ap_data,
    main = "ACF of AirPassengers Data",
    lag.max = 20,
    ylim = c(-1,1))
```




```r
pacf(ap_data,
     main = "PACF of AirPassengers Data",
     lag.max = 20,
     ylim = c(-1,1))
```




```r
# ---------------------------------------------------
# (j) Check stationarity using KPSS Test
# ---------------------------------------------------

kpss.test(ap_data)
```

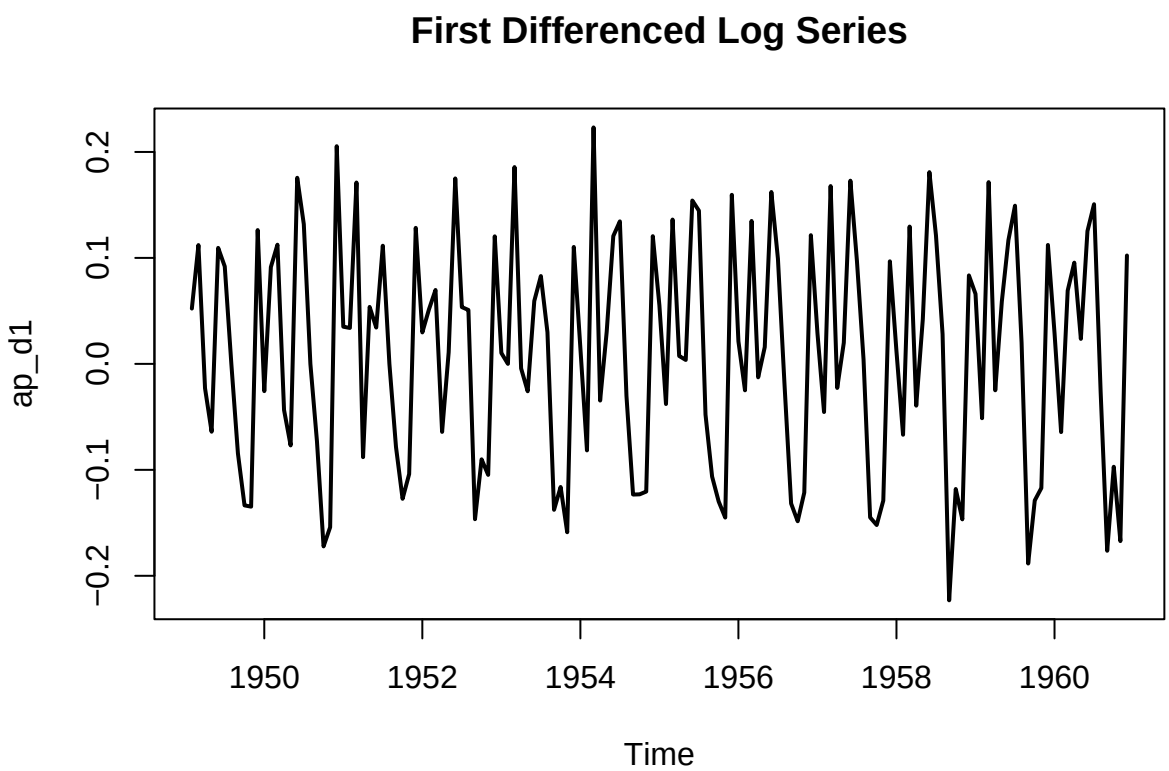
```
## Warning in kpss.test(ap_data): p-value smaller than printed p-value

##
##  KPSS Test for Level Stationarity
##
## data:  ap_data
## KPSS Level = 2.7395, Truncation lag parameter = 4, p-value = 0.01
```

```r
# ---------------------------------------------------
# (k) If non-stationary, make it stationary
# Use log transformation (variance stabilization)
# followed by differencing
# ---------------------------------------------------

ap_log <- log(ap_data)

ap_d1 <- diff(ap_log)

kpss.test(ap_d1)
```

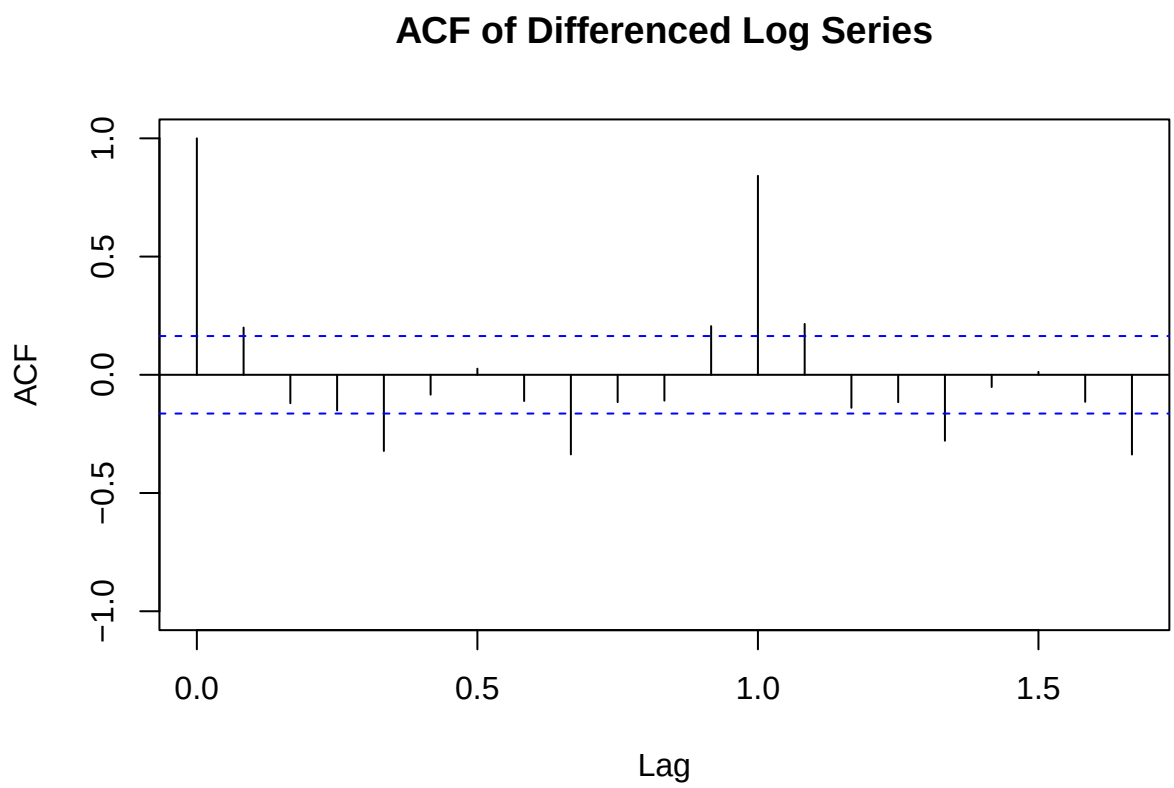
```
## Warning in kpss.test(ap_d1): p-value greater than printed p-value
```

```
##
##  KPSS Test for Level Stationarity
##
## data:  ap_d1
## KPSS Level = 0.028205, Truncation lag parameter = 4, p-value = 0.1
```

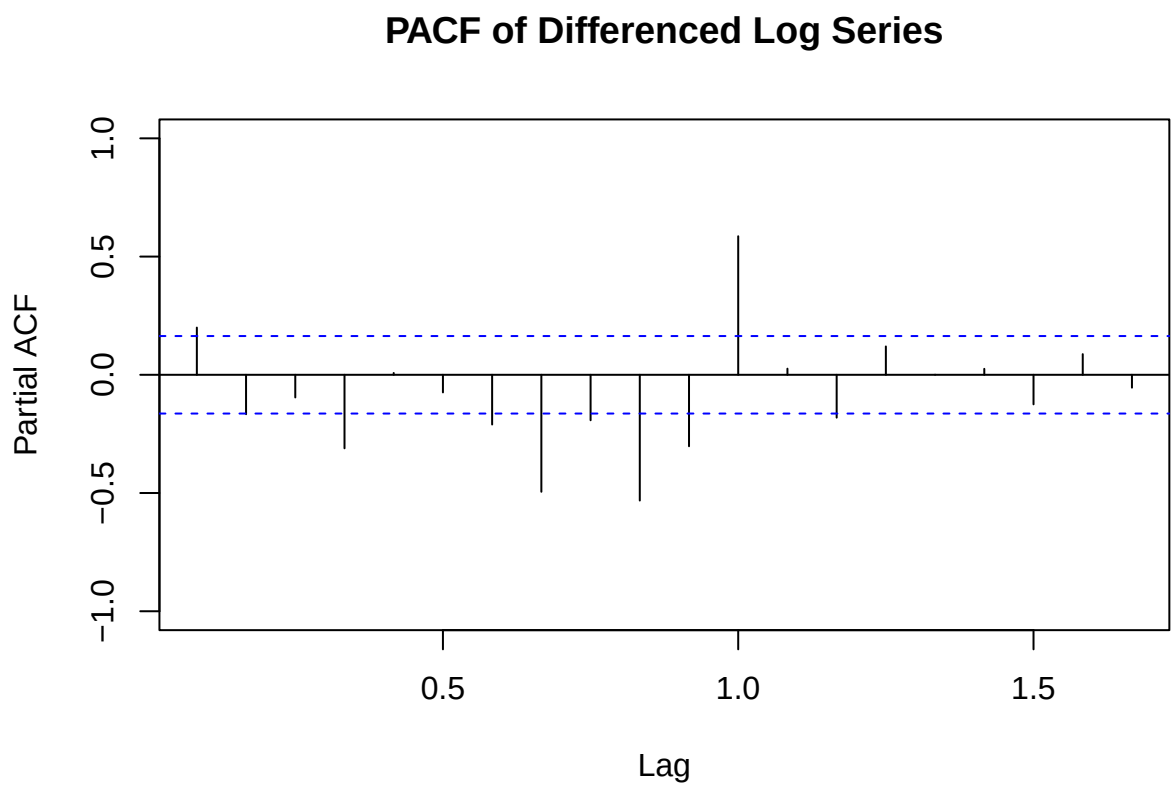
```r
plot(ap_d1,
     main = "First Differenced Log Series",
     xlab = "Time",
     lwd = 2)
```




```r
acf(ap_d1,
    main = "ACF of Differenced Log Series",
    lag.max = 20,
    ylim = c(-1,1))
```




```r
pacf(ap_d1,
     main = "PACF of Differenced Log Series",
     lag.max = 20,
     ylim = c(-1,1))
```

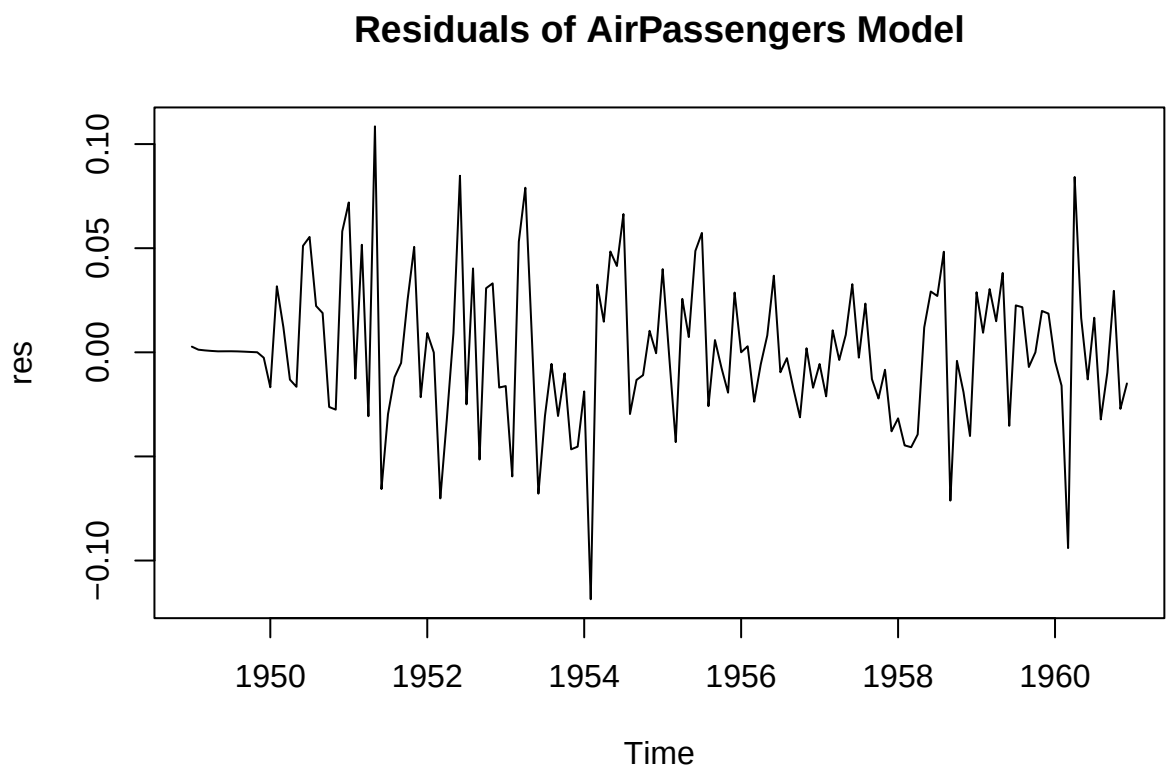



```r
# ---------------------------------------------------
# (l) Model Identification using ACF/PACF and AIC
# Since trend and seasonality are dominant,
# we use ARIMA model
# ---------------------------------------------------

ap_fit <- auto.arima(ap_log)

ap_fit
```

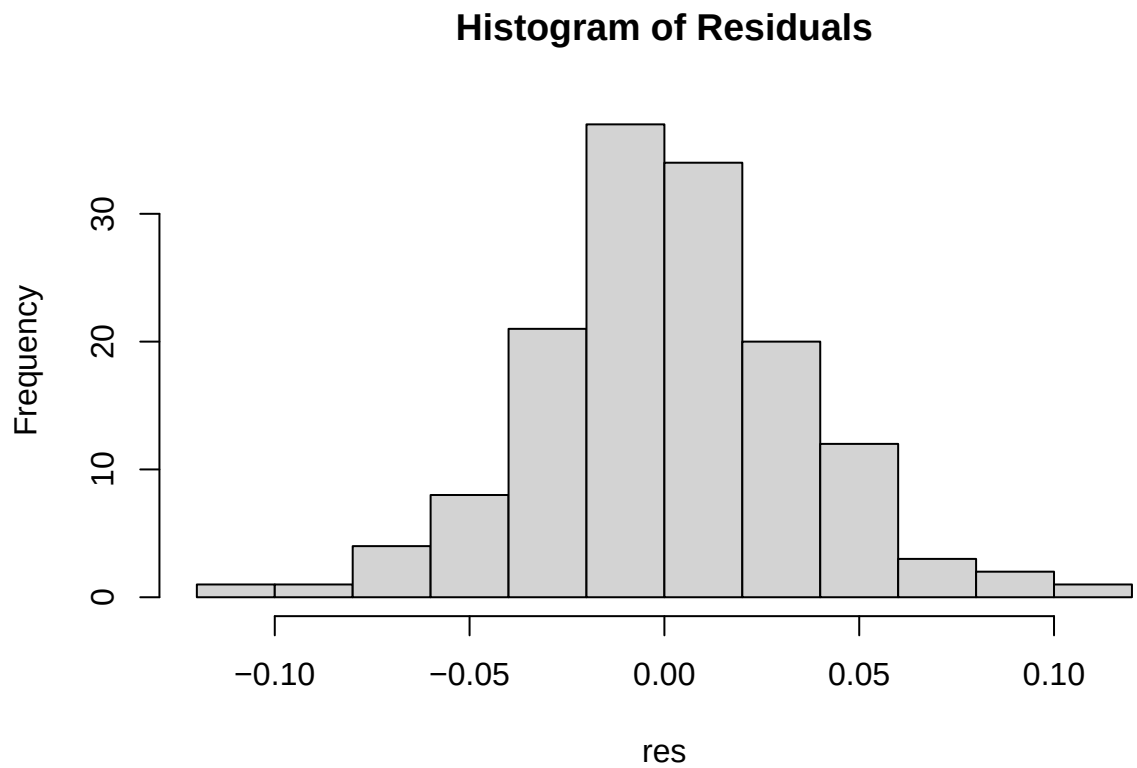
```
## Series: ap_log
## ARIMA(0,1,1)(0,1,1)[12]
##
## Coefficients:
##           ma1     sma1
##       -0.4018  -0.5569
## s.e.   0.0896   0.0731
##
## sigma^2 = 0.001371:  log likelihood = 244.7
## AIC=-483.4   AICc=-483.21   BIC=-474.77
```

```r
# ---------------------------------------------------
# (m) Fit the selected model and estimate parameters
# (Model fitted above using auto.arima)
# ---------------------------------------------------
```

```r
# ---------------------------------------------------
# (n) Check Goodness of Fit (Diagnostic Checking)
# ---------------------------------------------------

res <- ap_fit$residuals

# Plot residuals
plot(res,
     main = "Residuals of AirPassengers Model")
```




```r
# Check Normality
hist(res,
     main = "Histogram of Residuals")
```



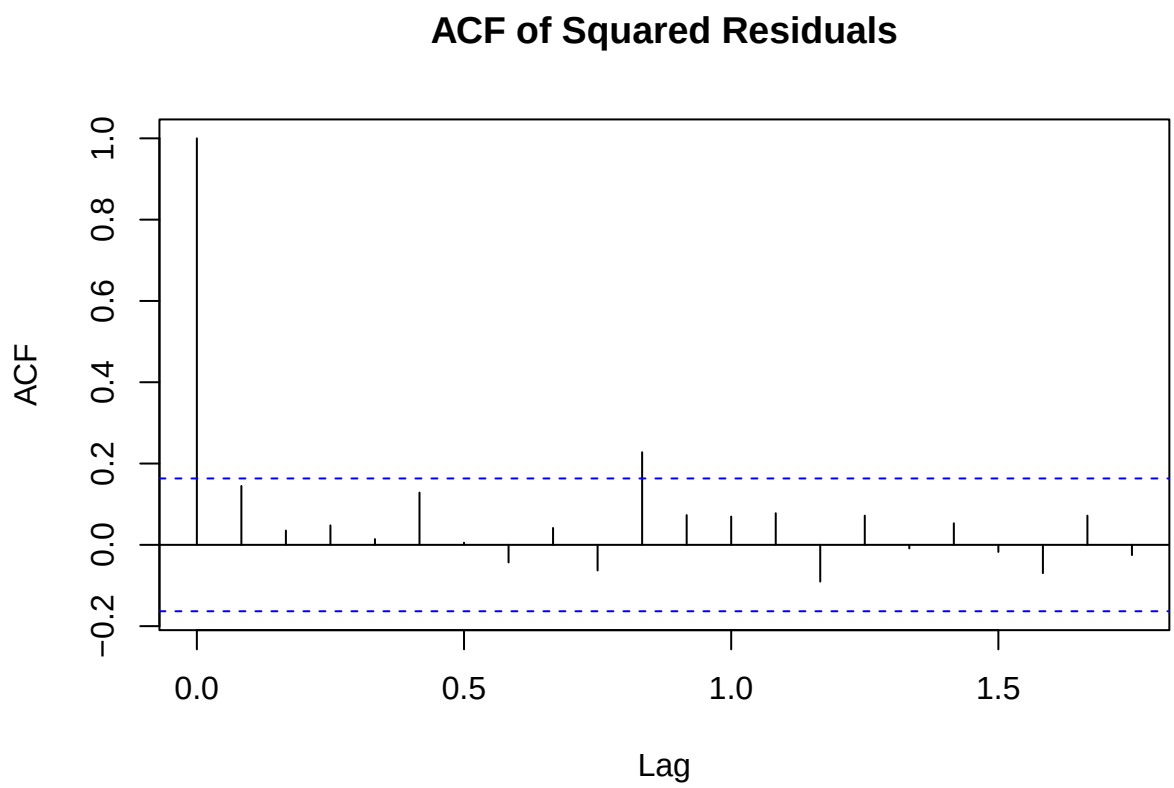

```r
shapiro.test(res)
```

```
##
##  Shapiro-Wilk normality test
##
## data:  res
## W = 0.98637, p-value = 0.1674
```

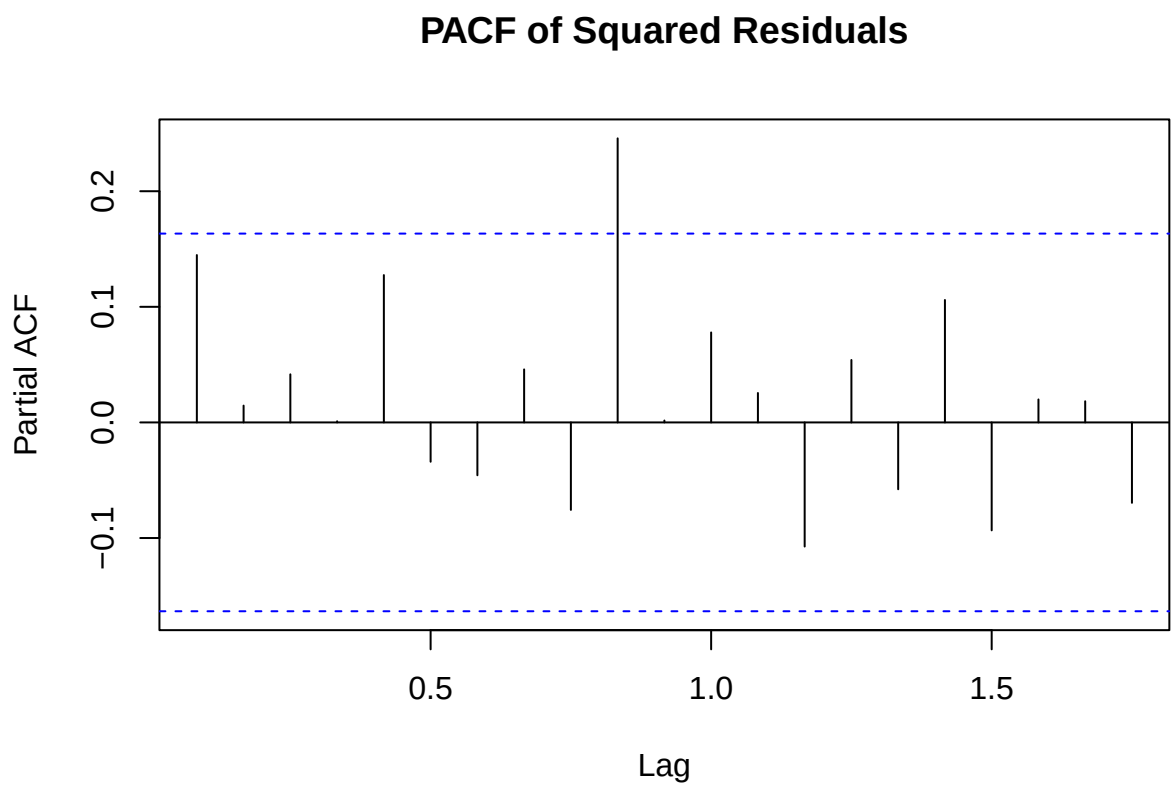
```r
# Check Serial Correlation
Box.test(res,
         lag = 20,
         type = "Box-Pierce")
```

```
##
##  Box-Pierce test
##
## data:  res
## X-squared = 16.062, df = 20, p-value = 0.7128
```

```r
# Check Heteroscedasticity
acf(res^2,
    main = "ACF of Squared Residuals")
```




```
pacf(res^2,
     main = "PACF of Squared Residuals")
```

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### Conclusion

The AirPassengers data shows a strong upward trend and multiplicative seasonality.

ACF and KPSS test confirmed that the original series is non-stationary.

After applying log transformation and first differencing, the series became stationary.

Based on trend and seasonal dominance, an ARIMA model was selected using AIC.

Diagnostic checking confirmed:

- Residuals are approximately normal
- No serial correlation
- Constant variance

Hence, the selected ARIMA model is a good fit for the AirPassengers data.